Applied Bayesian Statistics in Animal & Biological Sciences

2) (Hierarchical) linear prediction models

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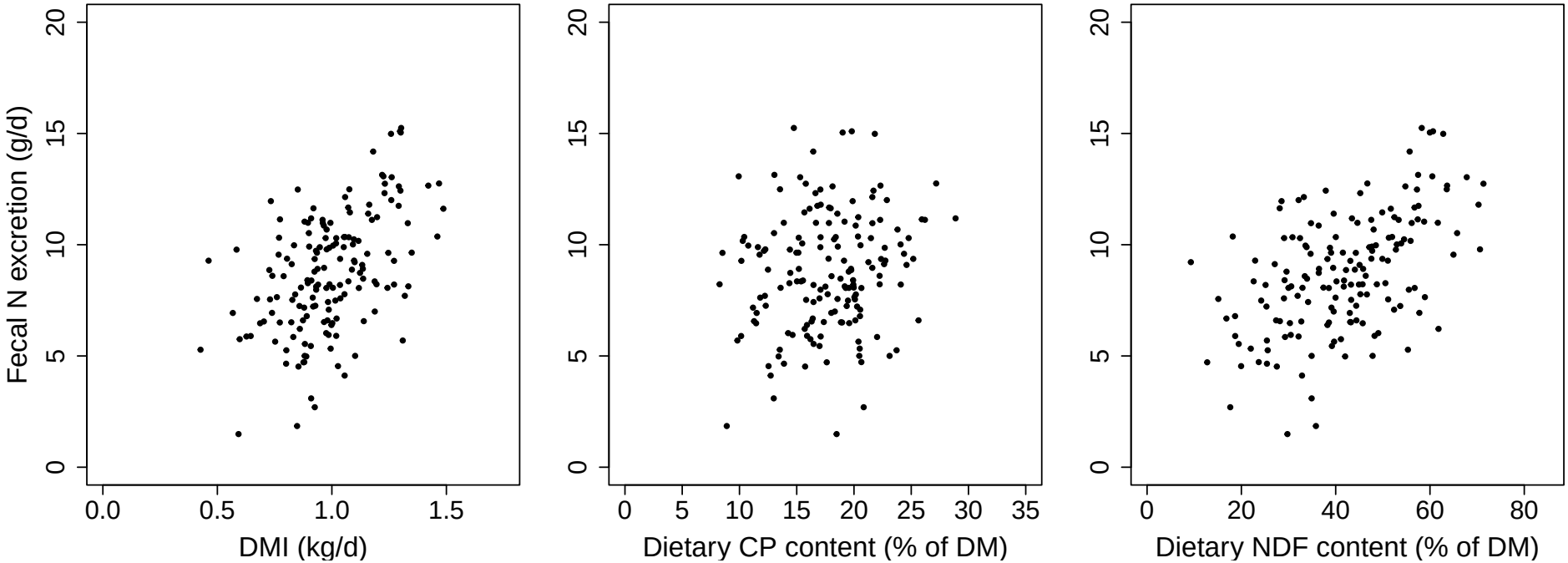
Today's program

- Prior distributions for linear models
- Build "fixed-effects" model STAN code and evaluate simulation output
- Assessing RSTAN output
- Build "mixed-effects" or hierarchical model STAN code and evaluate simulation output
- Consider vectorization in STAN code

Predicting fecal nitrogen excretion from sheep



$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \beta_3 x_{3,i} + \epsilon_i \quad (1)$$

$$\epsilon_i \sim N(0, \sigma_e^2) \quad (2)$$

Apply Bayes' theorem to linear models

For parameters that follow continuous distributions:

$$\overbrace{\underbrace{p(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{posterior}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{p(\boldsymbol{y})}}^{\text{Bayes' rule}} \propto \underbrace{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{prior times likelihood}} \tag{3}$$

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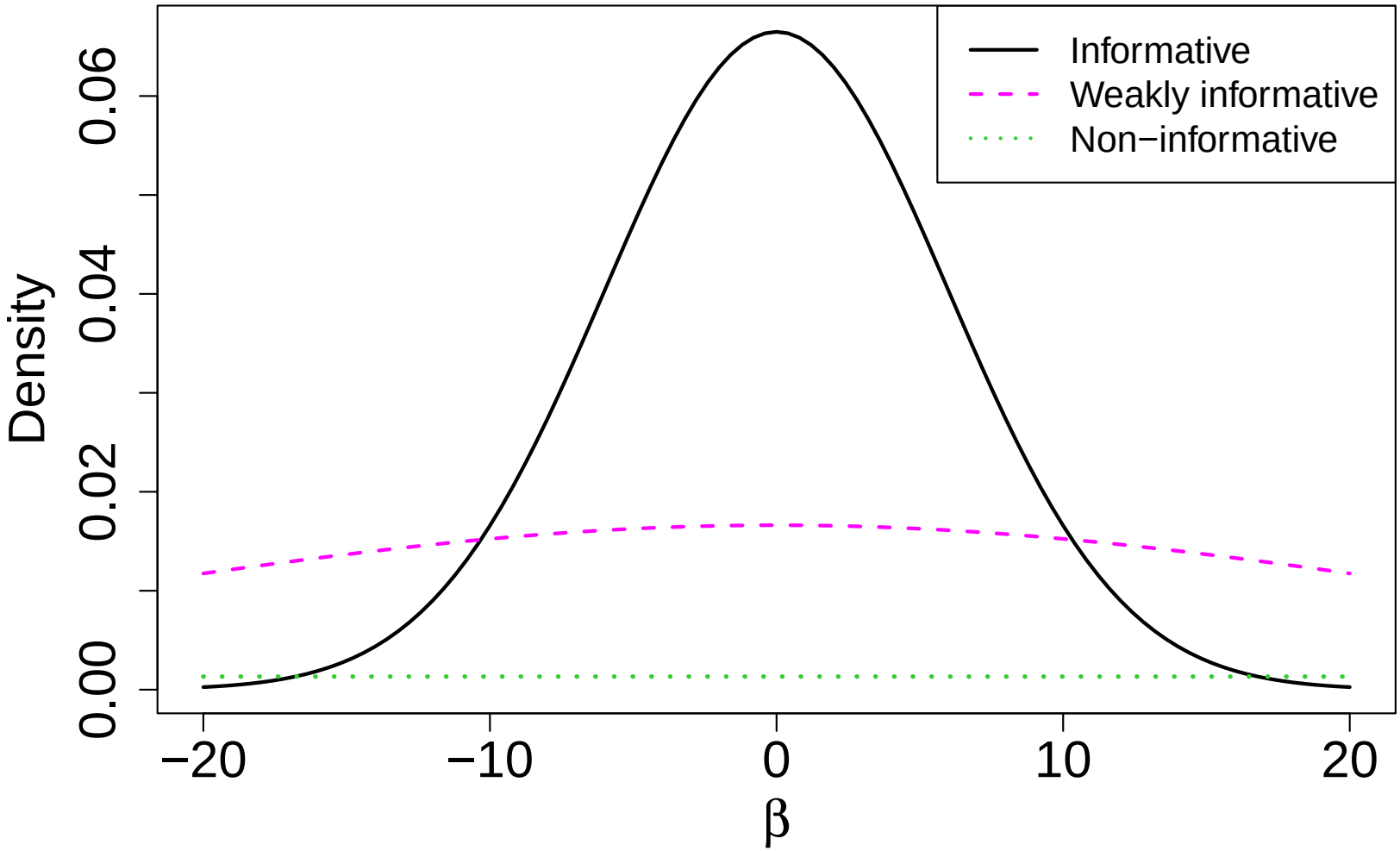
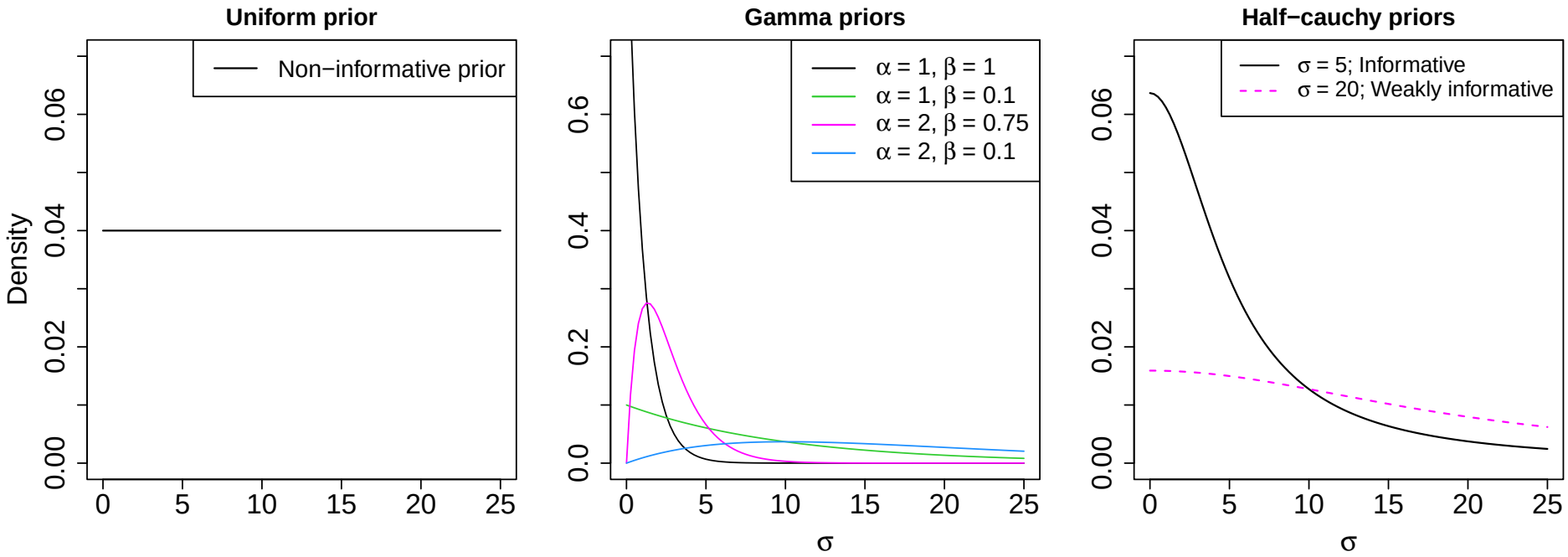
- Parameters are considered random variables following any distribution assigned
- Obtain posterior knowledge through optimizing $p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})$, i.e. use data and prior knowledge
- Powerful Markov-chain Monte Carlo algorithms used for optimization are run for n iterations from which we sample

Priors for linear prediction model - β parameters



$$\beta_0, \beta_1, \beta_2, \beta_3 \sim \{N(0, 6), N(0, 24), N(0, 300)\} \tag{4}$$

Priors for linear prediction model - σ parameter



$$\sigma_e \sim \text{Uniform}(0, 25); \text{ also called a flat prior} \quad (5)$$

$$\sigma_e \sim \text{Gamma}(\alpha, \beta); \text{ with } \alpha, \beta \text{ the shape and rate parameters} \quad (6)$$

$$\sigma_e \sim \text{half-cauchy}(\mu = 0, \sigma); \text{defined on } [\mu, \infty) \quad (7)$$

with μ, σ the location and scale parameters

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"Fixed-effects" linear model code in STAN

```
FN_FixedEf <- "
data {
  int<lower=0> N;                 // number of data points
  real y[N];                      // the Fecal nitrogen data
  real x_DMI[N];                    // the DMI data
  real x_CP[N];                    // the CP data
  real x_NDF[N];                    // the NDF data
} // End of data

parameters{
  real<lower=0> sigma_e;     // residual std dev
  real beta_0;               // regression coefficients
  real beta[3];               // regression coefficients
} // End of parameters

model {
  for(i in 1:N){
    y[i] ~ normal(beta_0 + beta[1]*x_DMI[i] + beta[2]*x_CP[i] + beta[3]*x_NDF[i], sigma_e);
  } // End of for loop i
  beta_0 ~ normal(0,300);
  beta ~ normal(0,300);
  sigma_e ~ cauchy(0,15);
} // End of model
```

- STAN is C++ based, note the semi-colons...
- STAN only takes integers and numeric values, and uses σ, not σ^2

RSTAN code to perform model simulation

```
### RSTAN code ###
library(rstan)
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)

### Initialize STAN code in R
FN_FixedEf_mod <- stan_model(model_code=FN_FixedEf,model_name="FN_FixedEf")

### Generate data input for STAN model as a list
n_sample <- 160; n_studies <- 10

set.seed(6708)
sheep_FN_dat <- data.frame(x_DMI = rnorm(n_sample, 0.977, 0.209), # DMI
                     x_CP = rnorm(n_sample, 17.7, 4.33),    # CP
                     x_NDF = rnorm(n_sample, 41.6, 13.3),    # NDF
                     s_s = rep(rnorm(n_studies, 0, sqrt(1.68)), each=n_sample/n_studies),
                     e_res = rnorm(n_sample, 0, sqrt(2.25)) )

sheep_FN_dat <- data.frame(sheep_FN_dat, Study=rep(1:n_studies, each=n_sample/n_studies),
                     y = -4.36 + 6.03*sheep_FN_dat$x_DMI + 0.109*sheep_FN_dat$x_CP +
                            0.115*sheep_FN_dat$x_NDF + sheep_FN_dat$s_s + sheep_FN_dat$e_res)

sheep_FN_dat$Study <- as.factor(sheep_FN_dat$Study)

sheep_FN_dat <- list(N=n_sample, y=sheep_FN_dat$y,
                      x_DMI=sheep_FN_dat$x_DMI, x_CP=sheep_FN_dat$x_CP, x_NDF=sheep_FN_dat$x_NDF)
```

RSTAN code to perform simulation - process output

```
### Run the HMC simulation of the STAN model
sheep.rs1 <- sampling(FN_FixedEf_mod, data=sheep_FN_dat, pars=c("beta_0","beta","sigma_e"),
                      iter=15e3, warmup=1e3, thin=25, chains=2)

### Write output to csv file
print(summary(sheep.rs1, pars=c("beta_0","beta","sigma_e"))$summary)
```

```
##            mean se_mean    sd   2.5%    25%    50%    75%  97.5%    n_eff  Rhat
## beta_0   -3.246   0.035 1.088 -5.320 -3.946 -3.306 -2.534 -1.085  944.527 0.999
## beta[1]   5.721   0.024 0.747  4.288  5.236  5.728  6.220  7.231  959.127 1.003
## beta[2]   0.120   0.001 0.036  0.048  0.096  0.119  0.144  0.191 1118.421 1.000
## beta[3]   0.100   0.000 0.011  0.078  0.093  0.100  0.108  0.121  953.045 0.999
## sigma_e   1.902   0.003 0.107  1.710  1.828  1.897  1.972  2.126 1031.951 0.999
```

```
write.csv(summary(sheep.rs1, pars=c("beta_0","beta","sigma_e"))$summary, "sheep_FN_FixedEf_out.csv")

### Save traceplot in pdf format
pdf("ModPars_vs_iterations.pdf", height=7, width=11)
traceplot(sheep.rs1, pars=c("beta_0","beta","sigma_e"))
dev.off()
```

- 2 Markov-chain Monte Carlo chains run for 15×10^3 iterations
- The first 1×10^3 iterations are the considered the warmup or burnin to let the chains converge
- Effective sample size approaches: $2 \cdot \frac{15 \times 10^3 - 1 \times 10^3}{25} = 1.12 \times 10^3$

Evaluate (R)STAN model output (posteriors)

```
print(summary(sheep.rs1, pars=c("beta_0","beta","sigma_e"))$summary)
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```
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- Output comprises mean, se of mean, sd and various quantiles of parameter values

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- Notice the effective samples sizes (`n_eff`)

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```

- Output comprises mean, se of mean, sd and various quantiles of parameter values
- Notice the effective samples sizes (`n_eff`)
- $\hat{R}$ assesses convergence of chains by comparing "between" and "within" chain variations for a parameter

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##             mean se_mean    sd   2.5%    25%    50%    75%  97.5%    n_eff  Rhat
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```
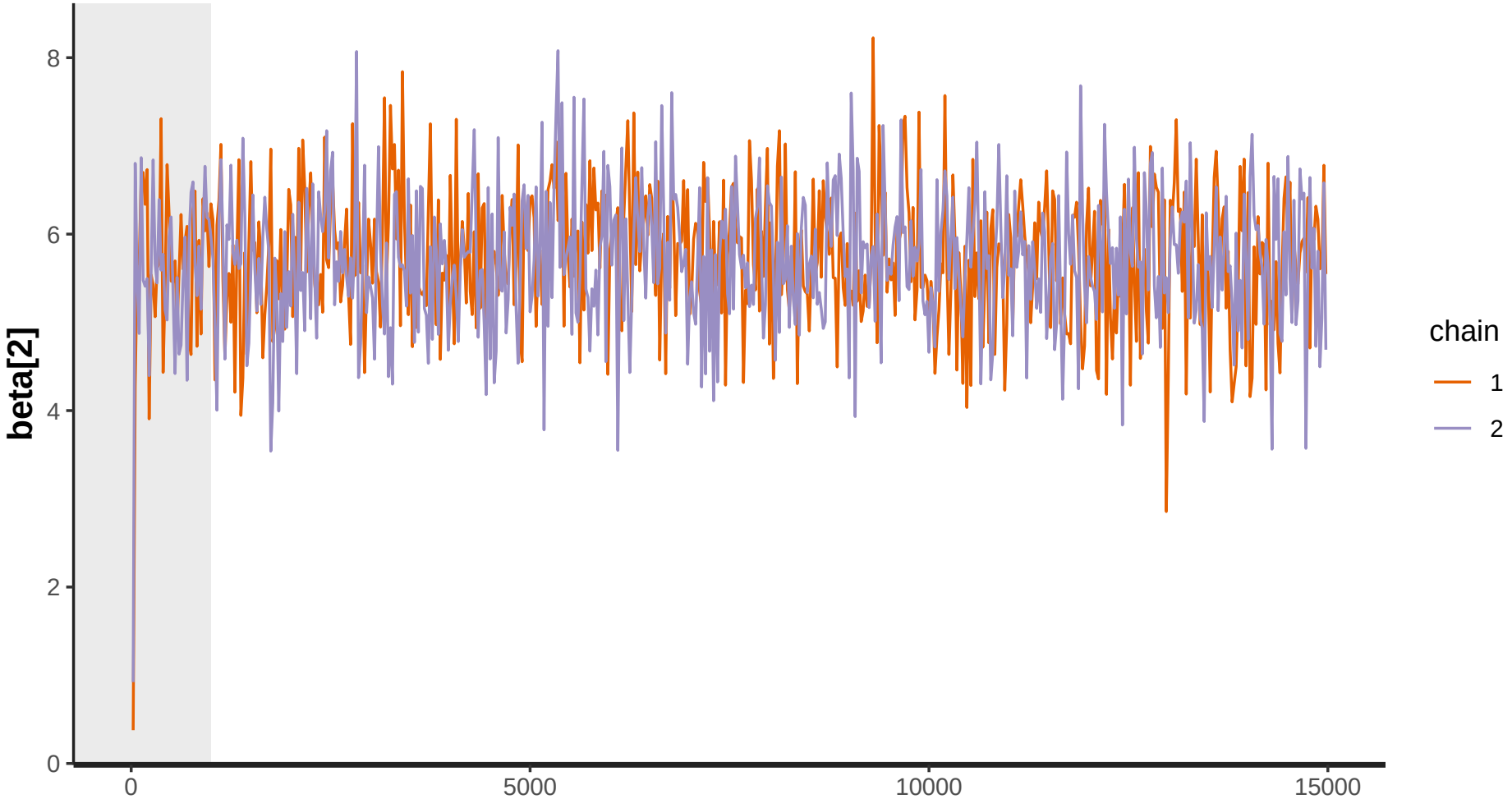
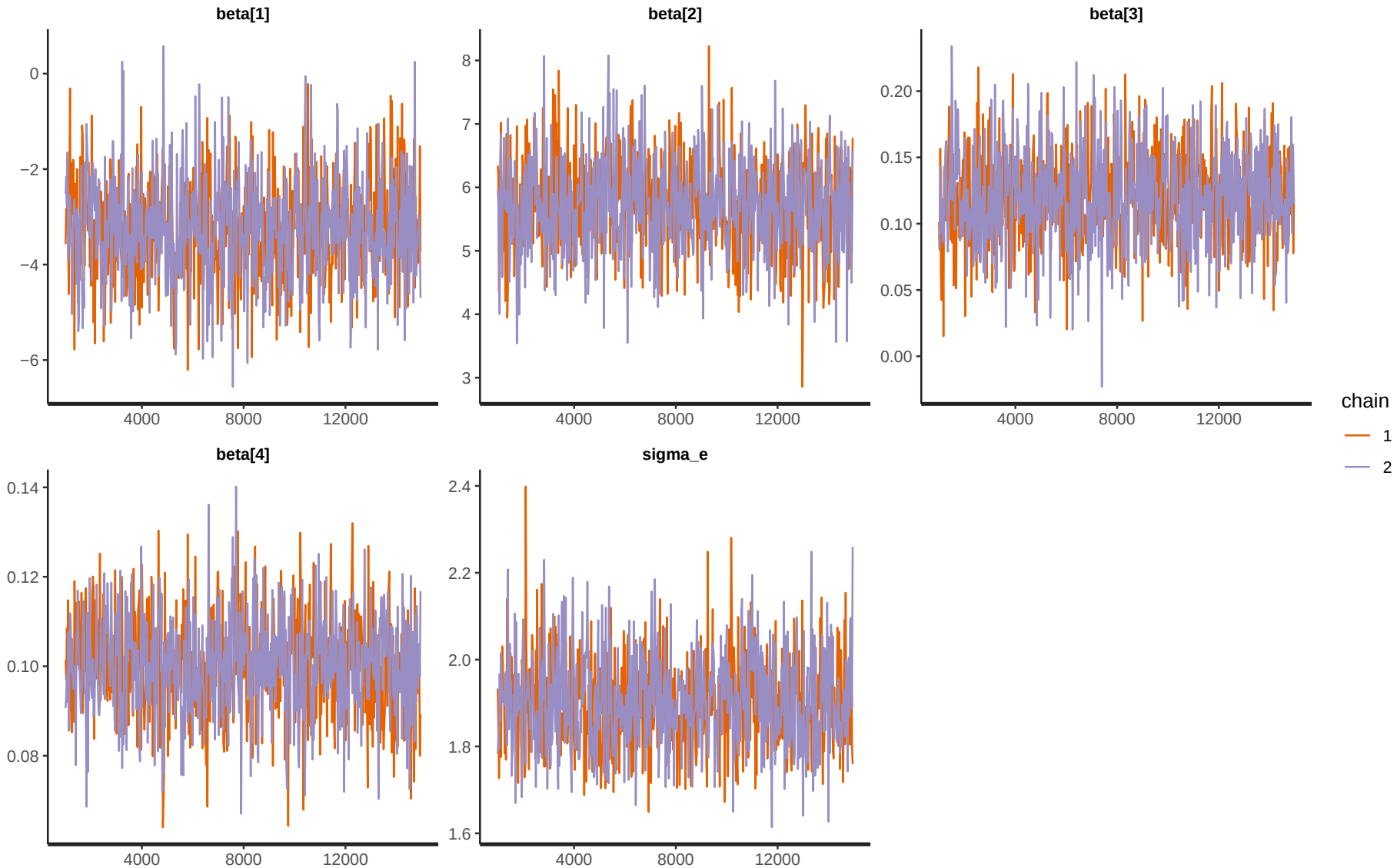

- Output comprises mean, se of mean, sd and various quantiles of parameter values
- Notice the effective samples sizes (`n_eff`)
- $\hat{R}$ assesses convergence of chains by comparing "between" and "within" chain variations for a parameter
- $\hat{R} \approx 1.00$, preferably $\hat{R} < 1.01$, certainly $\hat{R} < 1.05$

Model simulation in RSTAN - Traceplot

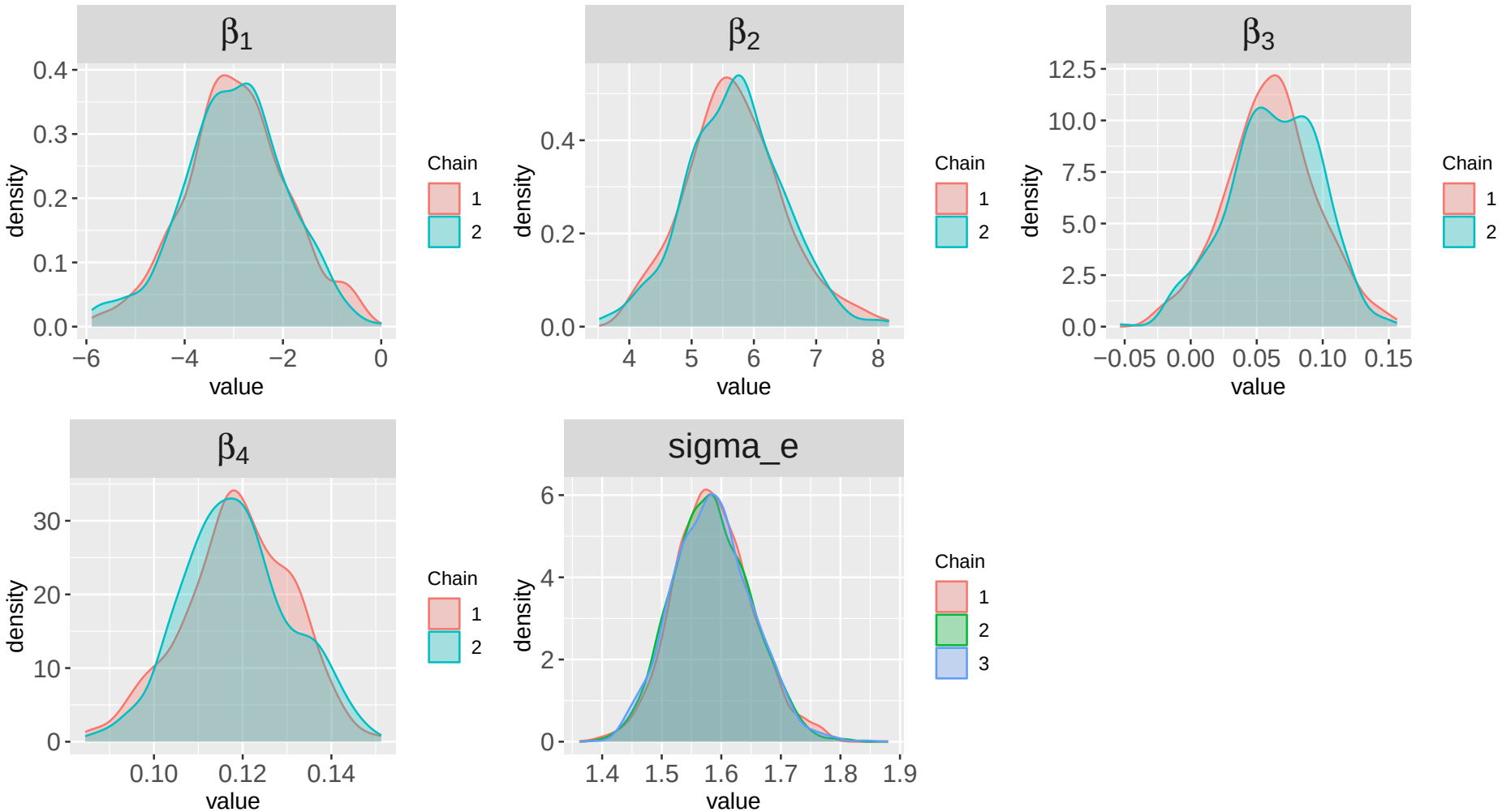


- First few iterations (burnin or warmup; here n =1000) are discarded

Model simulation in RSTAN - Traceplot

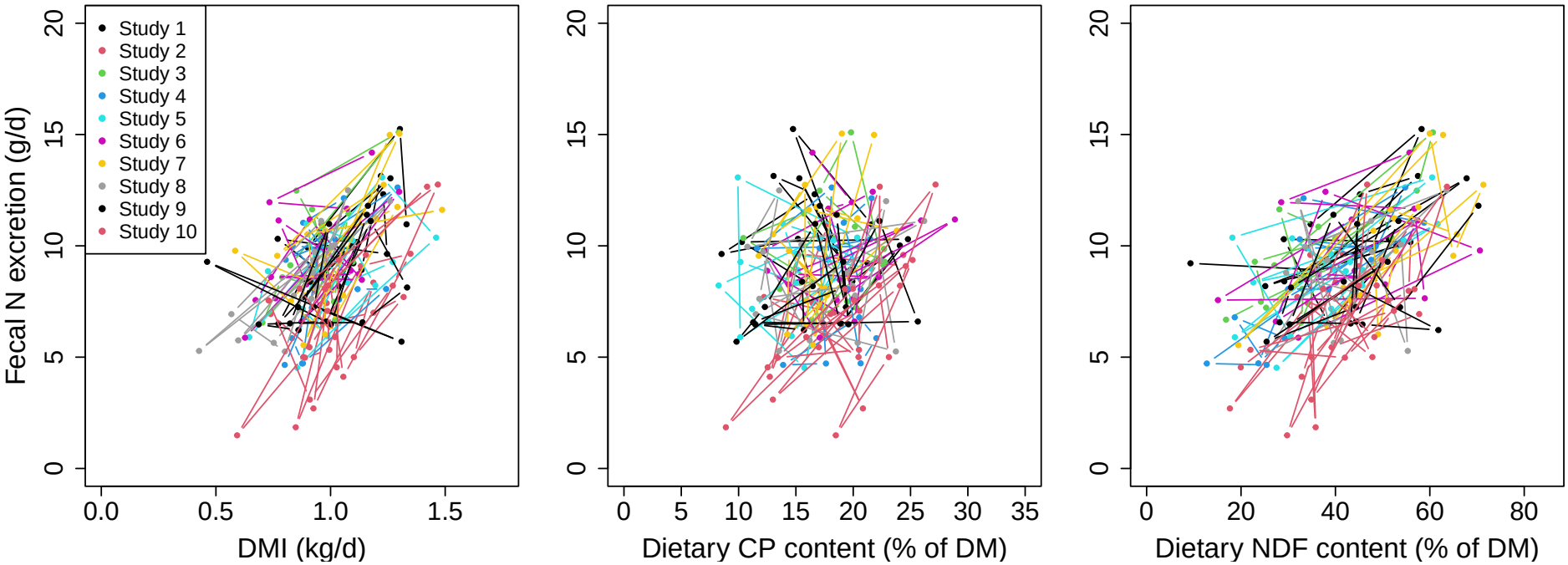


Model simulation in RSTAN - Posterior density plots



- Satisfied with the chains on these density plots?

Hierarchical or "Mixed-effects" modeling



$$y_{ij} = \beta_0 + \beta_1 x_{1,ij} + \beta_2 x_{2,ij} + \beta_3 x_{3,ij} + s_j + \epsilon_{ij} \tag{8}$$

$$s_j \sim N(0, \sigma_s^2) \tag{9}$$

$$\epsilon_{ij} \sim N(0, \sigma_e^2) \tag{10}$$

Hierarchical linear prediction model code in STAN

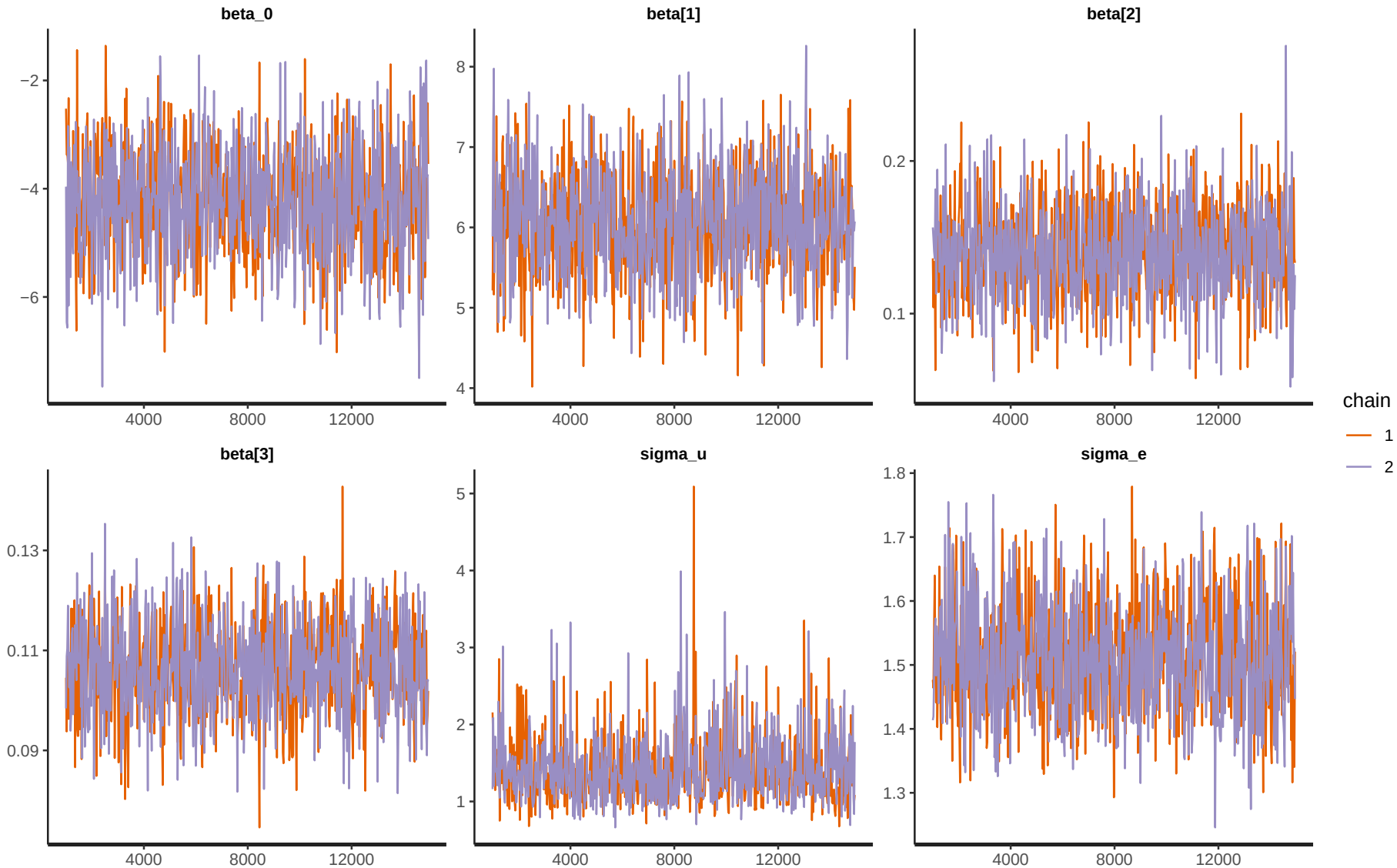
```
FN_MixedEf <- "
  data {
    int<lower=1> N;               // number of data points
    real y[N];                    // the FN data
    real x_DMI[N];                // the covariate 1 data
    real x_CP[N];                 // the covariate 2 data
    real x_NDF[N];                // the covariate 3 data
    int<lower=1> J;               // number of Studies
    int<lower=1, upper=J> EXP[N]; // the Studies
  } // end of data

parameters{
  real beta_0;             // regression coefficients
  real beta[3];            // regression coefficients
  real u[J];               // experiment intercepts
  real<lower=0> sigma_u;   // experiment std dev
  real<lower=0> sigma_e;   // residual std dev
} // end of parameters

model {
  for(i in 1:N){
    y[i] ~ normal(beta_0 + beta[1]*x_DMI[i] + beta[2]*x_CP[i] + beta[3]*x_NDF[i] + u[EXP[i]], sigma_e);
  } // end of for loop i

  beta_0 ~ normal(0,300); beta ~ normal(0,300)
  u ~ normal(0,sigma_u);
  sigma_u ~ cauchy(0,15); sigma_e ~ cauchy(0,15);
} // end of model
"
```

Hierarchical linear model RSTAN code

```
FN_MixedEf_mod <- stan_model(model_code=FN_MixedEf,model_name="FN_MixedEf")

### Extend the list data object with study information
sheep_FN_dat$J <- n_studies
sheep_FN_dat$EXP <- rep(1:n_studies, each=n_sample/n_studies)

### Run the HMC simulation of the STAN model
sheep.rs2 <- sampling(FN_MixedEf_mod, data=sheep_FN_dat, iter=15e3, warmup=1e3, thin=25, chains=2,
                      pars=c("beta_0","beta","u","sigma_u","sigma_e"))

print(summary(sheep.rs2, pars=c("beta_0","beta","u","sigma_u","sigma_e"))$summary)
```

```
##           mean se_mean    sd   2.5%    25%    50%    75%  97.5%    n_eff  Rhat
##beta_0  -4.222   0.030 1.000 -6.198 -4.892 -4.244 -3.523 -2.349 1079.988 1.000
##beta[1]  6.050   0.019 0.651  4.822  5.621  6.042  6.474  7.324 1161.076 1.000
##beta[2]  0.141   0.001 0.031  0.082  0.120  0.141  0.161  0.202 1300.157 1.001
##beta[3]  0.106   0.000 0.009  0.088  0.100  0.106  0.113  0.124 1103.222 0.999
##u[1]     0.960   0.017 0.573 -0.110  0.570  0.934  1.334  2.157 1091.671 1.000
##u[2]    -1.830   0.017 0.552 -2.937 -2.186 -1.839 -1.451 -0.746 1092.542 1.001
##u[3]     1.485   0.018 0.559  0.373  1.140  1.480  1.827  2.614 1009.265 1.000
##u[4]     0.388   0.016 0.556 -0.668  0.020  0.366  0.729  1.541 1221.293 0.999
##u[5]     0.654   0.018 0.593 -0.430  0.258  0.625  1.037  1.986 1084.186 0.999
##u[6]     0.809   0.017 0.570 -0.258  0.438  0.793  1.152  2.013 1121.052 1.000
##u[7]     0.013   0.017 0.561 -1.066 -0.355  0.011  0.363  1.196 1061.760 0.999
##u[8]     0.446   0.017 0.558 -0.718  0.098  0.443  0.809  1.582 1051.670 0.999
##u[9]    -0.687   0.018 0.576 -1.823 -1.064 -0.700 -0.312  0.463 1007.436 1.000
##u[10]   -1.994   0.017 0.584 -3.123 -2.388 -1.981 -1.612 -0.846 1174.574 0.999
##sigma_u  1.422   0.015 0.445  0.824  1.120  1.346  1.607  2.533  906.891 1.005
##sigma_e  1.511   0.003 0.089  1.350  1.451  1.505  1.568  1.697 1144.163 1.003
```

Hierarchical linear model traceplots



Vectorize linear model equation

Model equation in scalar and vectorized form

$$y_{ij} = \beta_0 + \beta_1 x_{1,ij} + \beta_2 x_{2,ij} + \beta_3 x_{3,ij} + s_j + \epsilon_{ij} \tag{11}$$

$$y_{ij} = \boldsymbol{x}_{ij}^T \boldsymbol{\beta} + s_j + \epsilon_{ij} \tag{12}$$

"Random-effects" of the model

$$s_j \sim N(0, \sigma_s^2) \tag{13}$$

$$\epsilon_{ij} \sim N(0, \sigma_e^2) \tag{14}$$

- Vectorizing in STAN speeds up the MCMC simulation and shortens runtime

Vectorized linear model STAN code

```
FN_MixedEf_X <- "
  data {
    int<lower=0> N;                 // number of data points
    real y[N];                      // the FN data
    int<lower=1> K;                 // number of predictors
    matrix[N,K+1] x;                // the N covariate dataframe
    int<lower=1> J;                 //number of experiments
    int<lower=1, upper=J> EXP[N]; // the experiments
  } // end of data

parameters{
  vector[K+1] beta;          // slope rcs
  vector[J] u;               // experiment intercepts
  real<lower=0> sigma_u;     // experiment std dev
  real<lower=0> sigma_e;     // residual std dev
  } // end of parameters

model {
  y ~ normal(x*beta + u[EXP], sigma_e);

  beta ~ normal(0,300);
  u ~ normal(0,sigma_u);
  sigma_u ~ cauchy(0,15);
  sigma_e ~ cauchy(0,15);
  } // end of model
"
```

Vectorized linear model RSTAN code

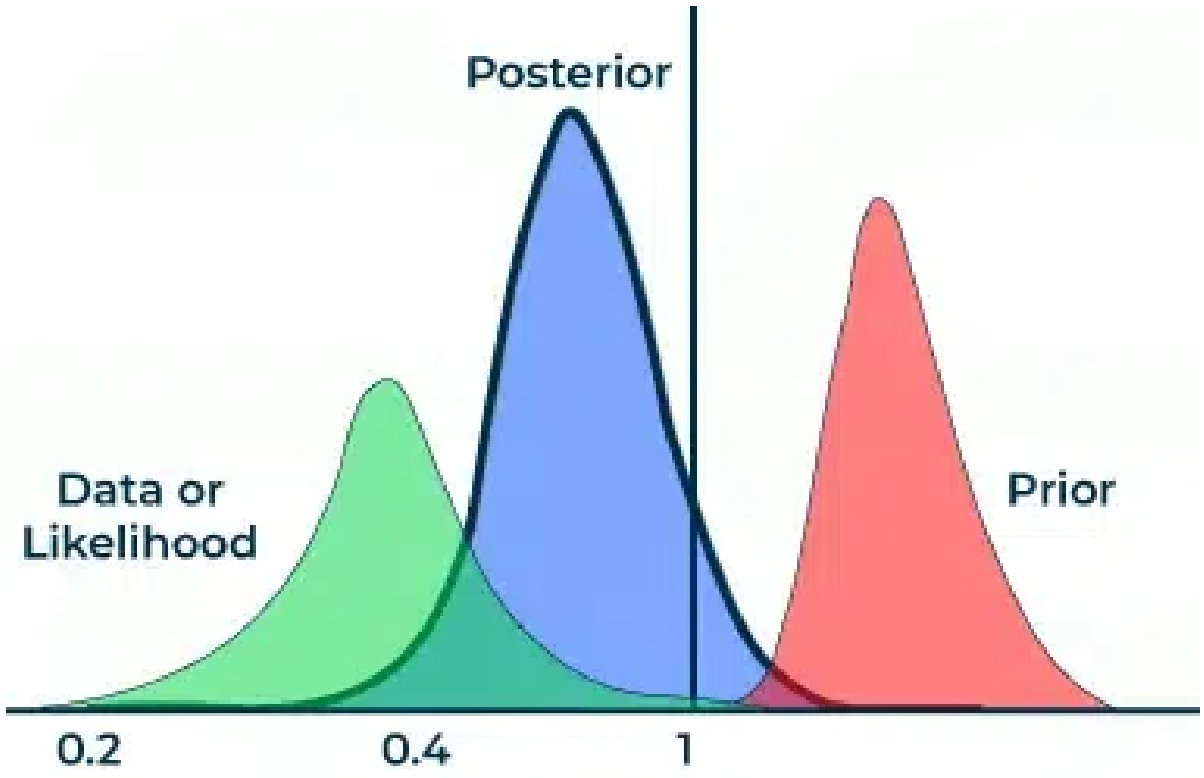
```
FN_MixedEf_X_mod <- stan_model(model_code=FN_MixedEf_X,model_name="FN_MixedEf_X")

### Extend the list data object with study information
sheep_FN_dat_X <- list(N=sheep_FN_dat$N, y=sheep_FN_dat$y, K=3, J=sheep_FN_dat$J, EXP=sheep_FN_dat$EXP,
                       x=data.frame(x_0=1, DMI=sheep_FN_dat$x_DMI,
                                    CP=sheep_FN_dat$x_CP, NDF=sheep_FN_dat$x_NDF))

# Note first column of data.frame object x
print(head(sheep_FN_dat_X$x))
```

```
##   x_0      DMI        CP       NDF
## 1   1 1.099482 21.264705  9.288876
## 2   1 1.135212 19.741308 36.469310
## 3   1 1.219533 13.050128 57.486815
## 4   1 1.229884 16.637352 45.220294
## .   . .        .         .
```

```
### Run the HMC simulation of the STAN model
sheep.rs3 <- sampling(FN_MixedEf_X_mod, data=sheep_FN_dat_X, pars=c("beta","u","sigma_u","sigma_e"),
                      iter=15e3, warmup=1e3, thin=25, chains=2)

print(summary(sheep.rs3, pars=c("beta","sigma_u","sigma_e"))$summary)
```

```
##           mean se_mean    sd   2.5%    25%    50%    75%  97.5%    n_eff  Rhat
##beta[1] -4.166   0.029 1.036 -6.315 -4.830 -4.129 -3.457 -2.174 1237.621 1.000
##beta[2]  6.023   0.020 0.657  4.695  5.610  6.028  6.467  7.292 1084.055 1.000
##beta[3]  0.141   0.001 0.031  0.082  0.120  0.140  0.162  0.201 1072.481 0.999
##beta[4]  0.106   0.000 0.010  0.087  0.100  0.106  0.113  0.125 1099.593 1.001
##sigma_u  1.414   0.015 0.435  0.803  1.118  1.335  1.615  2.550  892.255 0.999
##sigma_e  1.511   0.003 0.088  1.355  1.450  1.506  1.567  1.695  899.006 1.000
```

Thank you for your attention!



Questions?

Exercise lecture 2

- Take the code from the slides and reproduce the model output
- Increase/decrease the number of iterations and chains. Then check mean and standard deviations of posteriors
- Check the run times of the scalar and vectorized model codes
- Play around with priors, both distributions and parameters, and evaluate your output
- Bonus: extend this univeriate model to a bivariate prediction model for methane emissions and N excretion
- Bonus: write a linear model that uses Reproducing Kernel Hilbert Spaces: `https://www.ncbi.nlm.nih.gov/books/NBK583975/` or `https://doi.org/10.1534/genetics.114.164442`